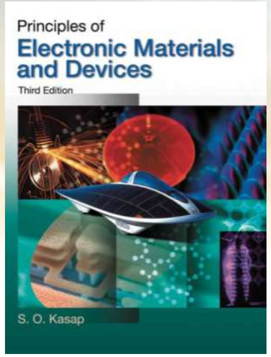
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》
Solid-State Electronics



Ch1-4
Elementary Materials Science Concepts

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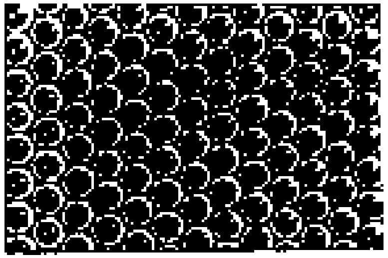
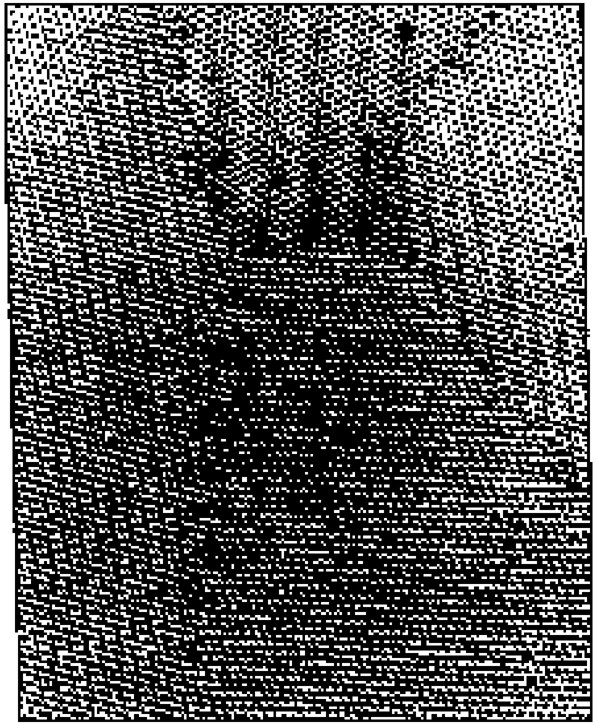
Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

1.9 Crystalline Defects 晶体中的缺陷

Defect：The difference of atomic arrangement between an ideal crystal and an actual crystal is called a crystal defect.



ideal crystal



actual crystal with defects

Ideal crystals

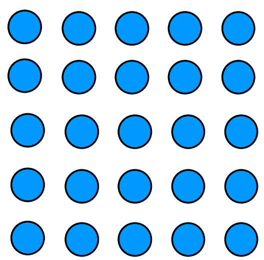
The particles are arranged in strict accordance with the spatial lattice.

Actual crystals

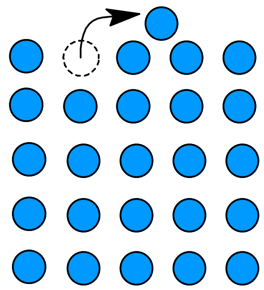
There are various structural incompleteness.

Crystalline defects 晶体缺陷

Above the absolute zero temperature, all crystals have atomic vacancies or atoms missing from lattice sites in the crystal structure. The vacancies exist as a requirement of thermal equilibrium and are called **thermodynamic defects** （热力学缺陷）



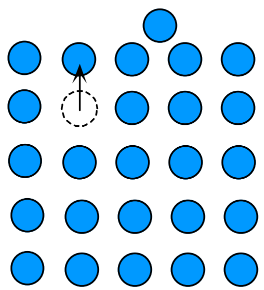
(a) Perfect crystal without vacancies



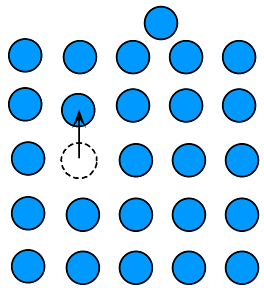
(b) An energetic atom at the surface breaks bonds and jumps on to a new adjoining position on the surface. This leaves behind a vacancy.

Vacancy concentration
空位浓度：

$$n_v = N \exp\left(-\frac{E_v}{kT}\right)$$



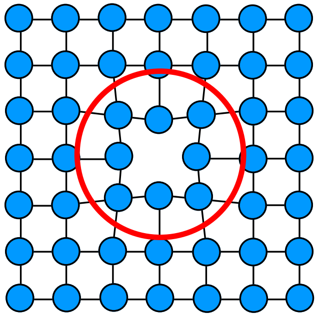
(c) An atom in the bulk diffuses to fill the vacancy thereby displacing the vacancy towards the bulk.



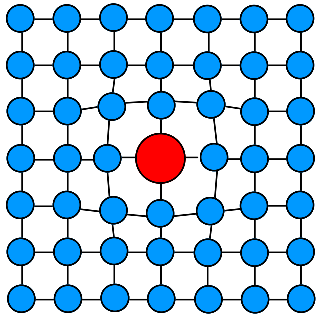
(d) Atomic diffusions cause the vacancy to diffuse into the bulk.

1.9.1 Point defect 点缺陷

1) Vacancies: There should be original vacancies at the lattice node location. This defect is called “vacancy”.



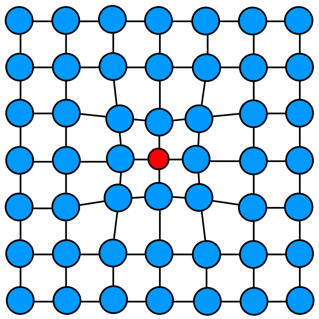
(a) A vacancy in the crystal.



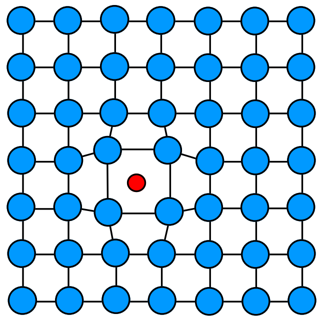
(b) A substitutional impurity in the crystal. The impurity atom is larger than the host atom.

(b),(c)
Substitutional impurity
置换型杂质

2) Displacement defects: impurity atoms replace original atoms.



(c) A substitutional impurity in the crystal. The impurity atom is smaller than the host atom.



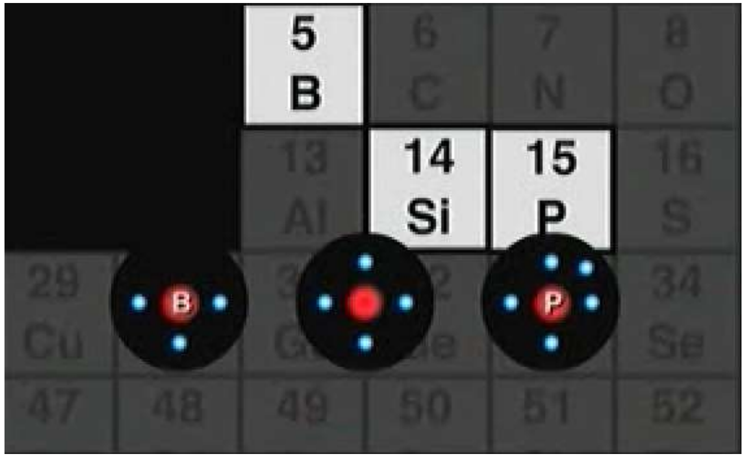
(d) An interstitial impurity in the crystal. It occupies an empty space between host atoms.

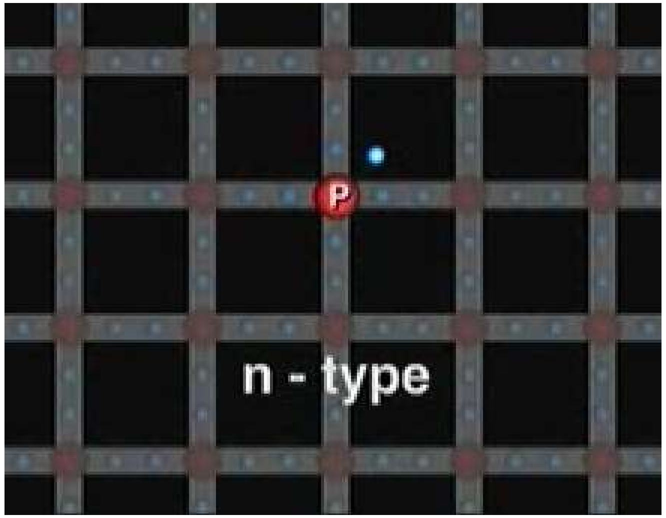
(d)
Interstitial impurity
填隙杂质

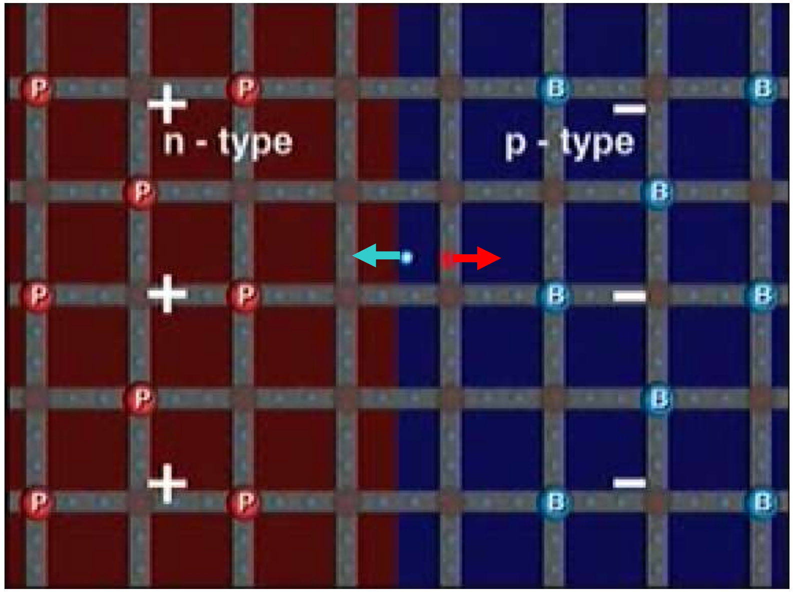
3) Interstitial atoms: In the lattice non-junction position, it is often the gap of the crystal lattice, and extra atoms appear.

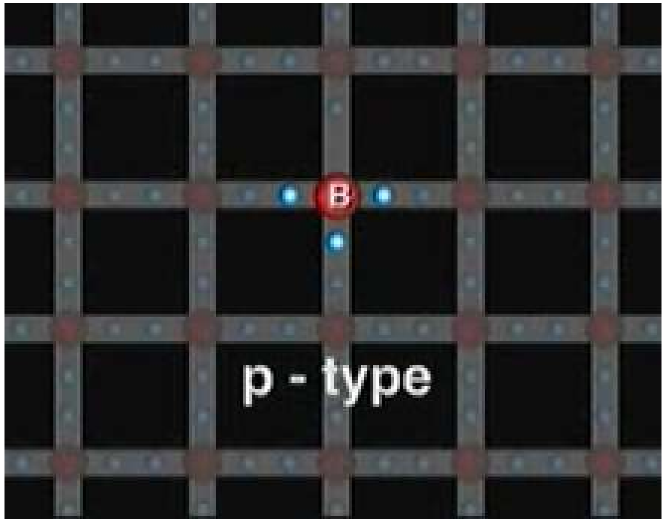
Point defects in the crystal structure. The regions around the point defect become distorted; the lattice becomes strained.

Doping of semiconductor

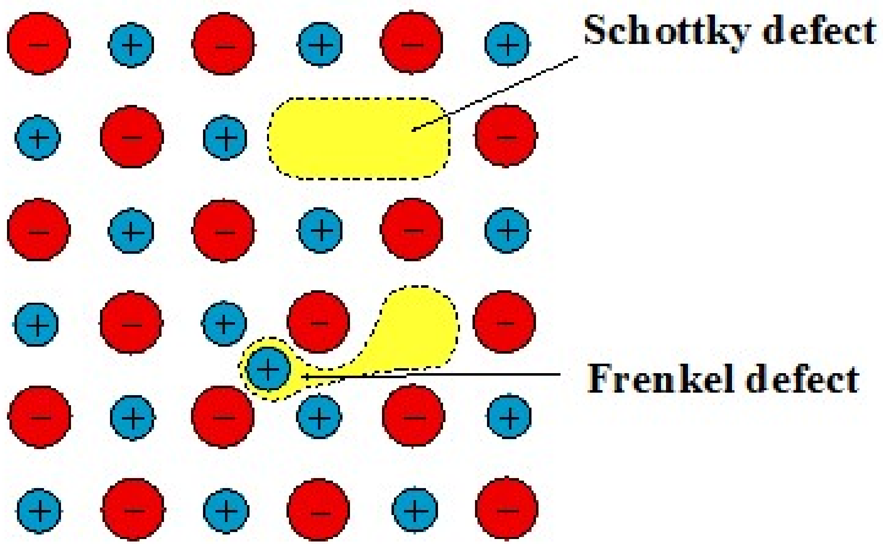








Point defect 点缺陷



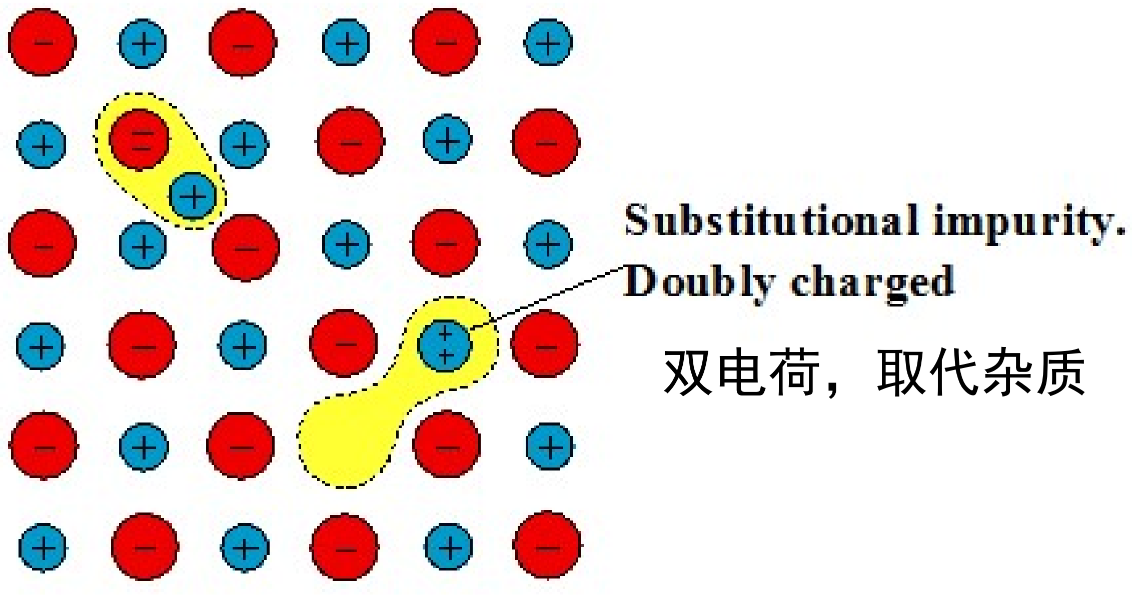
Schottky defect （肖特基缺陷）： missing a pair of cation-anion

- The internal atoms migrate to the surface leaving a vacancy at the original point. Vacancies can help ion diffusion, increase conductivity, and have certain optical and electrical properties.
- Disadvantages: new atomic layer is added to the crystal surface, and there are only vacancy defects inside the crystal, and the crystal volume expands and the density decreases.

Frenkel defect （弗伦克尔缺陷）： constituted by the interstitial ion and the vacancy

- Due to thermal fluctuations, an atom jumps from a normal point to a gap position, producing a gap and a filling atom.
- Vacancies and interstitial atoms appear simultaneously. The crystal volume does not change, and the crystal does not change in density due to vacancies.

Point defect 点缺陷



if Mg^{2+} replaces Na^{+}
→Na^{+} vacancy or one more Cl^{-}

if O^{2-} replaces Cl^{-}
→ one more cation or less Cl^{-}

Ionic crystals can have substitutional and interstitial impurities, the ionic crystal must be neutral.
保持电中性；离子晶体缺陷类型取决于离子晶体的成分，相对尺寸以及电荷数。

Impurity and Defect

- Defects resulting from the introduction of additional impurities are also referred to as constituent defects.
- The concentration of impurity defects is independent of temperature
- In order to purposefully improve device performance, artificially introduce impurity atoms

For example

1) Semiconductor like Silicon

$$10^5 \text{ Si atoms} \xrightarrow{\text{doped one B atom}} \text{conductivity increased by } 10^3 \text{ times}$$

2) Ruby laser

$$\text{crystal } Al_2O_3 \xrightarrow[\text{amount of Cr+}]{\text{doped with small}} \text{forming a luminescent center}$$

Influence of point defects on material properties

Mechanism

Regardless of the existence of the point defect, the atoms will be slightly deviated from the original position, that is, the lattice distortion at small area is caused.

Influences

1) Change the resistance: the flow directional of electrons is subjected to unbalanced forces (traps) around point defects, increasing the resistance; in ionic crystals, vacancies can help ion diffusion and increase conductivity.

2) Accelerate the diffusion of atoms: vacancies can serve as a intermediate station for atomic motion.

3) Other crystal defects are formed: supersaturated vacancies can concentrate to form internal voids, and the collapse of a concentrated one forms dislocations.

4) Changing the optical properties of a material: changes the color of the material.

VACANCY CONCENTRATION IN A METAL The energy of formation of a vacancy in the aluminum crystal is about 0.70 eV. Calculate the fractional concentration of vacancies in Al at room temperature, 300 K, and very close to its melting temperature 660 °C. What is the vacancy concentration at 660 °C given that the atomic concentration in Al is about 6.0×10^{22} cm^{-3}?

EXAMPLE 1.15

SOLUTION

Using Equation 1.35, the fractional concentration of vacancies are as follows:
At 300 K,

$$\frac{n_v}{N} = \exp\left(-\frac{E_v}{kT}\right) = \exp\left[-\frac{(0.70\text{ eV})(1.6 \times 10^{-19}\text{ J eV}^{-1})}{(1.38 \times 10^{-23}\text{ J K}^{-1})(300\text{ K})}\right]$$
$$= 1.7 \times 10^{-12}$$

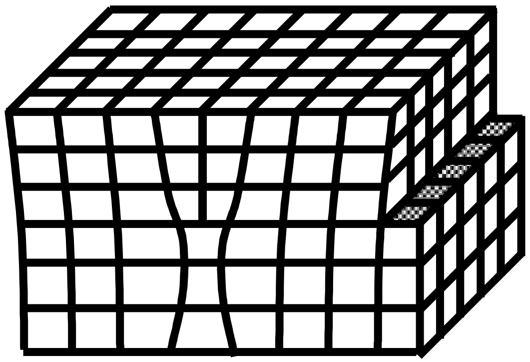
At 660 °C or 933 K,

$$\frac{n_v}{N} = \exp\left(-\frac{E_v}{kT}\right) = \exp\left[-\frac{(0.70\text{ eV})(1.6 \times 10^{-19}\text{ J eV}^{-1})}{(1.38 \times 10^{-23}\text{ J K}^{-1})(933\text{ K})}\right]$$
$$= 1.7 \times 10^{-4}$$

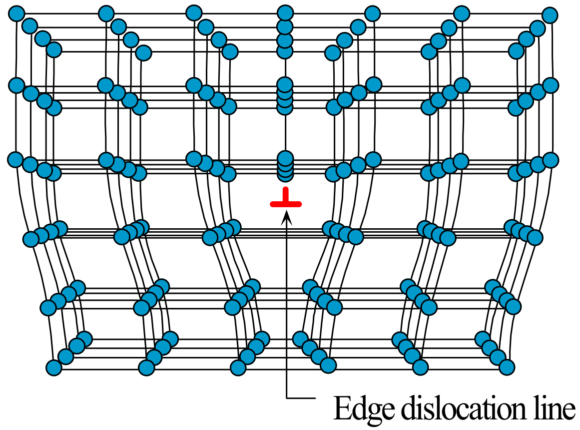
That is, almost 1 in 6000 atomic sites is a vacancy. The atomic concentration N in Al is about 6.0×10^{22} cm^{-3}, which means that the vacancy concentration n_v at 660 °C is

$$n_v = (6.0 \times 10^{22}\text{ cm}^{-3})(1.7 \times 10^{-4}) = 1.0 \times 10^{19}\text{ cm}^{-3}$$

1.9.2 Line defects: edge and screw dislocations

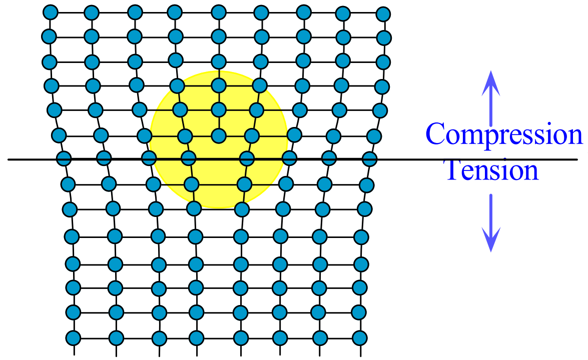


When there is a edge dislocation in the crystal, like chopping wood, half of the crystal is inserted into the upper half of the crystal, and is called the "edge dislocation".



(a) Dislocation is a line defect. The dislocation shown runs into the paper.

edge dislocation
刃位错

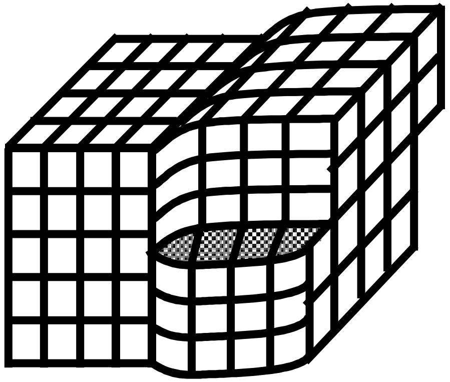


压应力
张应力

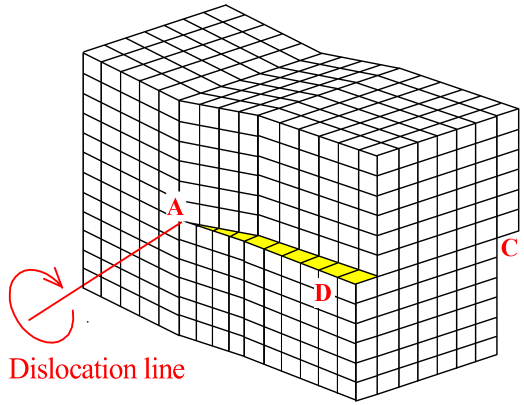
(b) Around the dislocation there is a strain field as the atomic bonds have been compressed above and stretched below the islocation line

Dislocation in a crystal is a line defect which is accompanied by lattice distortion and hence a lattice strain around it.

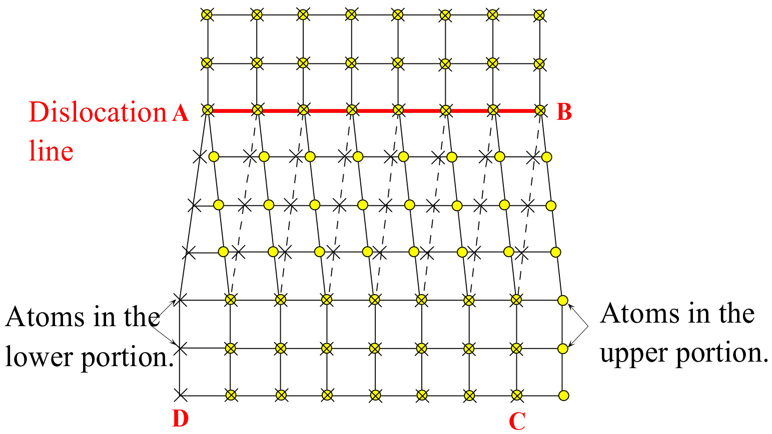
Screw Dislocation 螺位错



crystal with screw dislocation



(a) A screw dislocation in a crystal.

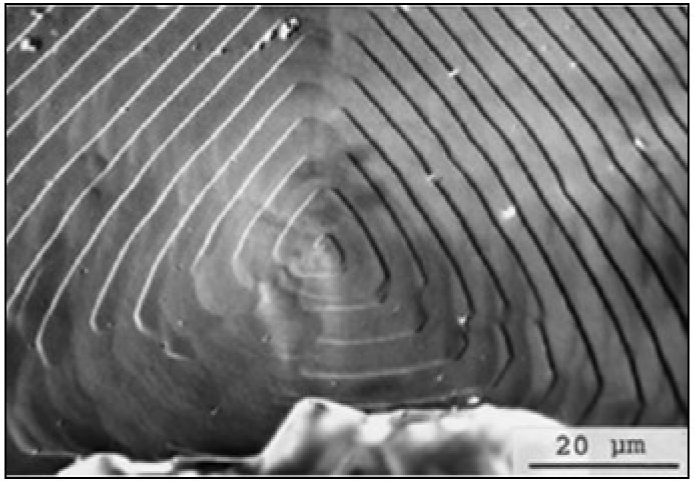


(b) The screw dislocation in (a) as viewed from above.

A spiral surface similar to a spiral step is formed at the original crystal face, and the position of the rotation axis is the screw dislocation.

Under the external force, the upper part of the crystal is moved backward by one atomic distance, and a spiral surface is formed in the boundary. The boundary line is the screw dislocation.

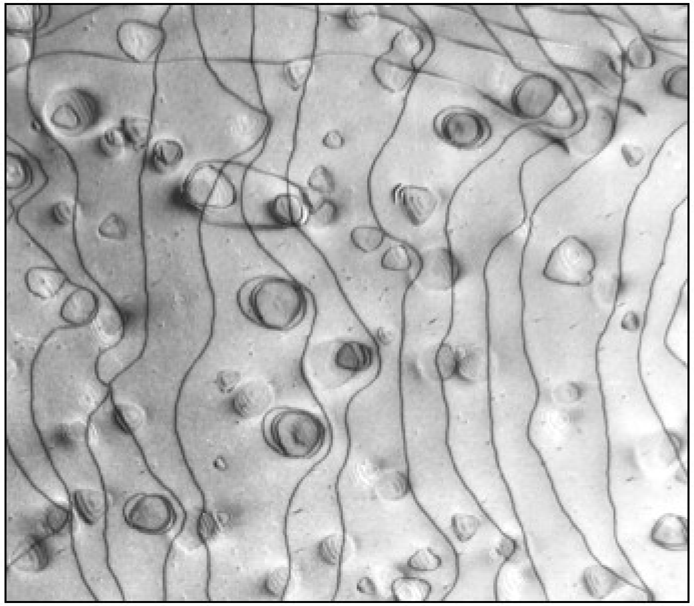
A screw dislocation involves shearing one portion of a perfect crystal with respect to another portion on one side of a line (AB).



dislocation lines can be observed through transmission electron microscopy (TEM)

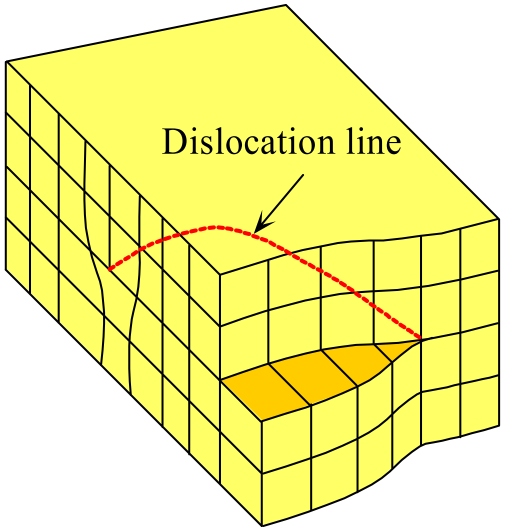
The photograph of the surface of a synthetic diamond grown on the (111) surface of natural diamond from sodium carbonate solvent at 5.5 GPa and 1600 °C.

SOURCE: Courtesy of Dr. Hisao Kanda, National Institute for Materials Science, Ibaraki, Japan.

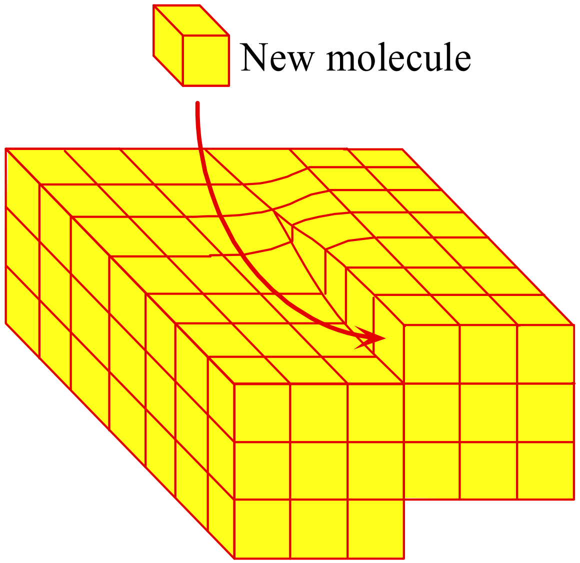


Dislocations can be seen by examining a thin slice of the sample under a transmission electron microscope (TEM). They appear as dark lines and loops as shown here in a Ni–Si alloy single crystal. The loop dislocations are around Ni_3Si particles inside the crystal. The sample had been mechanically deformed, which generates dislocations.

SOURCE: Courtesy of Professor John Humphreys, UMIST, England. (J. Humphreys and V. Ramaswamy in *High Voltage Electron Microscopy*, ed. P. R. Swann. C. J. Humphreys and M. J. Goringe, New York: Academic Press, 1974, p. 26.)



A mixed dislocation.

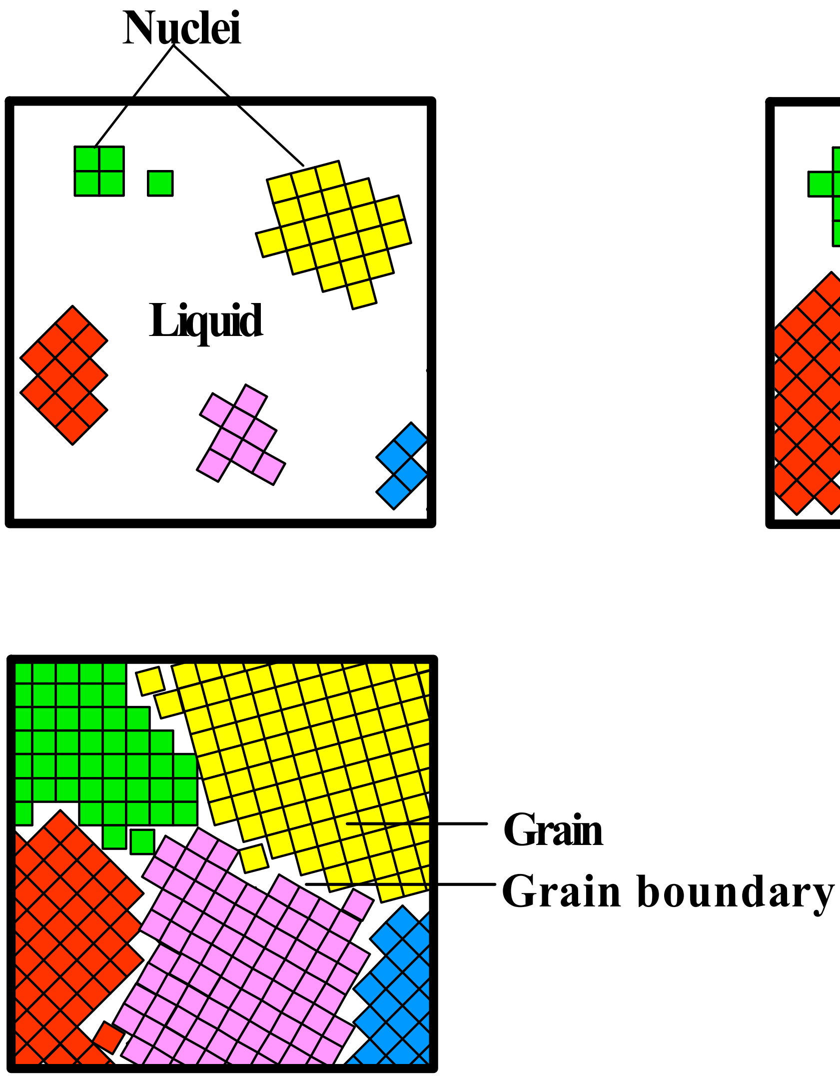


Growth spiral on the surface of a polypropylene crystal due to screw dislocation aided crystal growth.

SOURCE: Photo by Phillip Geil, Courtesy of Case Western Reserve University.

Far away from the dislocation, the atomic arrangement is close perfect arrangement. The generation and movement of line defects are closely related to the toughness （韧性） and brittleness （脆性） of the material.

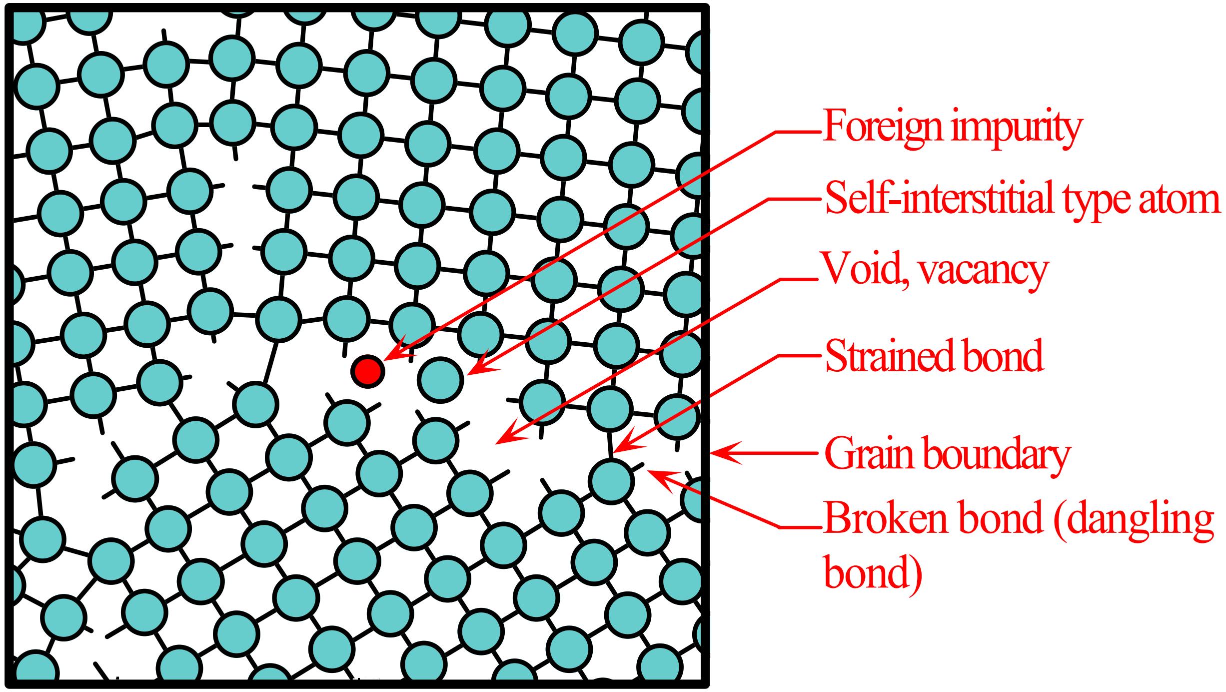
1.9.3 Planar defect: grain boundaries



The atoms can diffuse more easily along a grain boundary.

- Less bonds need to be broken due to the presence of voids
- the bonds are strained and easily broken

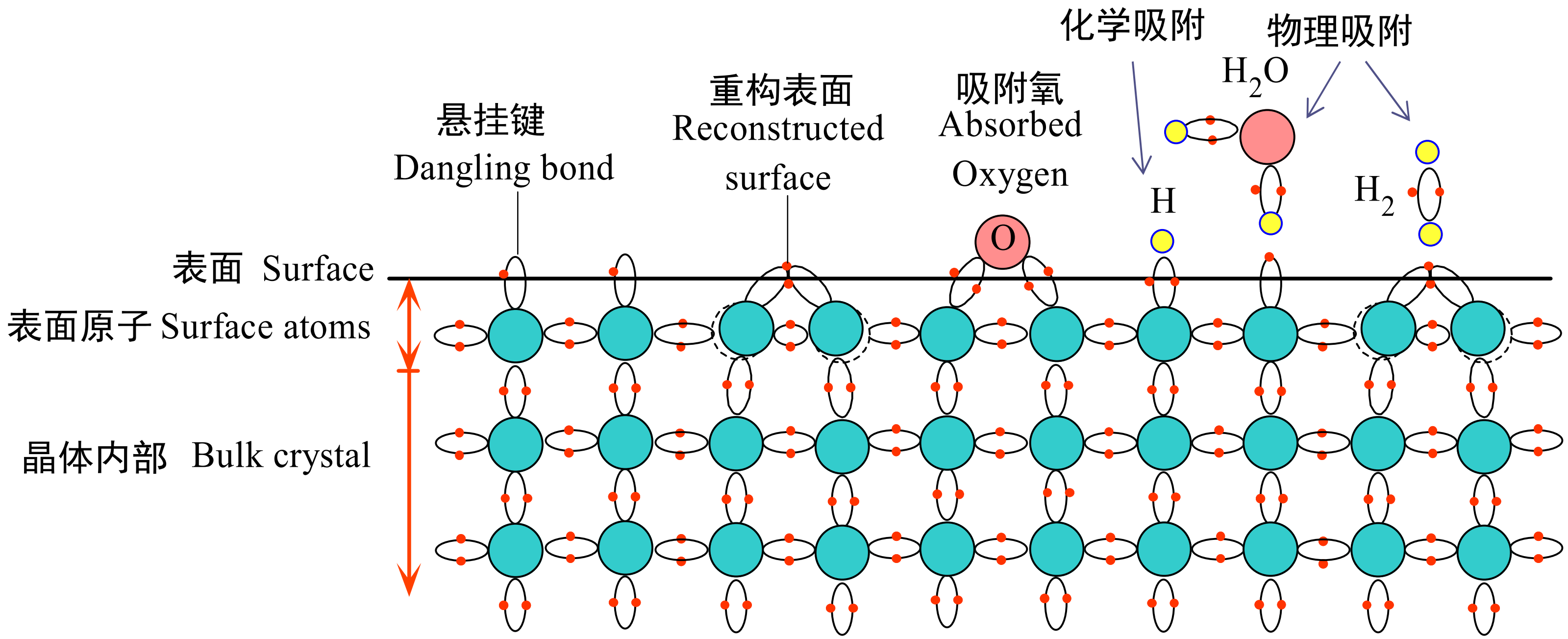
Planar defect: grain boundaries



晶界处存在断键，空隙，空位，受应变的键和填隙型原子。晶界处的结构是无序的，晶界处原子能量高于晶粒内部原子的能量

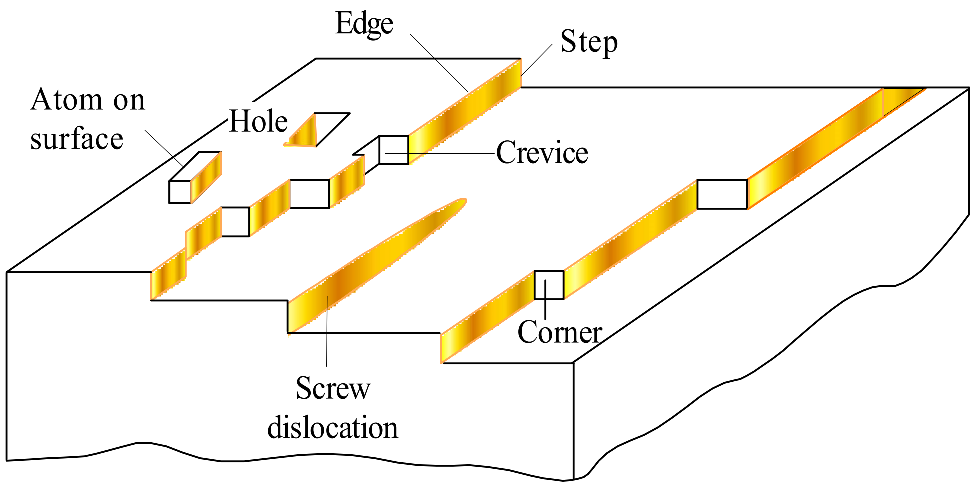
The grain boundaries have broken bonds, voids, vacancies, strained bonds and "interstitial" type atoms. The structure of the grain boundary is disordered and the atoms in the grain boundaries have higher energies than those within the grains.

1.9.4 Crystal surfaces and surface properties 晶体表面和表面性质



At the surface of a hypothetical two dimensional crystal, the atoms cannot fulfill their bonding requirements and therefore have broken, or dangling, bonds. Some of the surface atoms bond with each other; the surface becomes reconstructed. The surface can have physisorbed and chemisorbed atoms.

在假想二维晶体表面的原子不能满足成键要求，因此存在断键或悬挂键。一些表面原子同其他原子键合，是表面重构。表面可有存在物理吸附和化学吸附的原子。

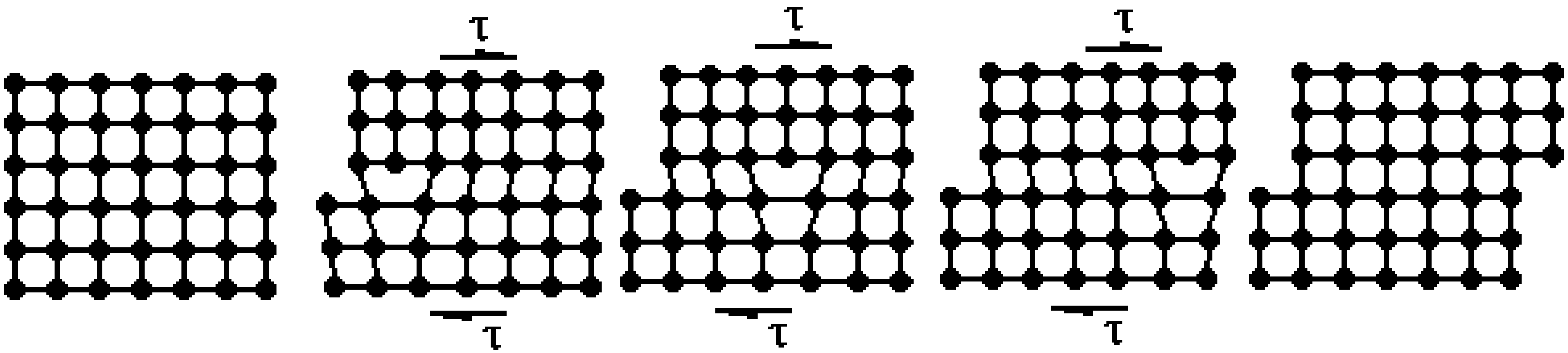


Typically a crystal surface has many types of imperfections such as steps, ledges, kinks, crevices, holes and dislocations.

晶体表面具有很多不完整性，如台阶，边，弯曲，裂口，孔洞及错位。

The surface structure depends greatly on the mode of surface formation, which invariably involves thermal and mechanical processing, and previous environmental history.

Slip of Edge Dislocation 刃位错的滑移

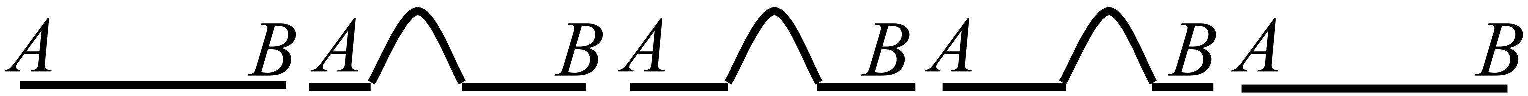


The atomic structure near the dislocation line has been significantly distorted, causing the atom to be in an unstable state. If applying a small shear force, and the distorted atom will move parallel along the shear force direction on the slip surface. When the dislocation line is removed, An atomic step is formed on the surface of the crystal.

位错线附近原子结构已有明显畸变，使原子处于不稳定状态，施加较小的切变力，畸变后的原子将在滑移面上平行于切变力方向移动；当位错线移出，在晶体表面形成一个原子台阶。

Analogy: The movement of a hump

类比：　地毯鼓包的移动



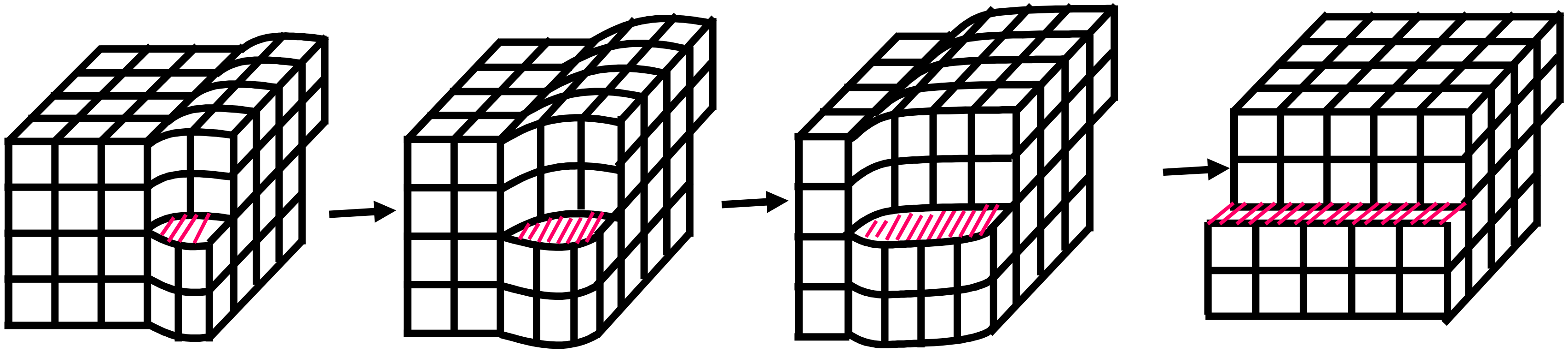
Make a hump on one side of the carpet, and it is easier to move the hump from one side to the other then moving the whole carpet

使地毯一边隆起一个鼓包，鼓包从一边移动到另一边比地毯整块移动所花的力气要小得多

Dislocations are similar to an "atomic carpet" hump that helps atomic slip

位错就类似一个“原子地毯”的鼓包，帮助原子滑移

Slip of Screw Dislocation 螺位错的滑移



Similar like tear the paper （类比撕纸）

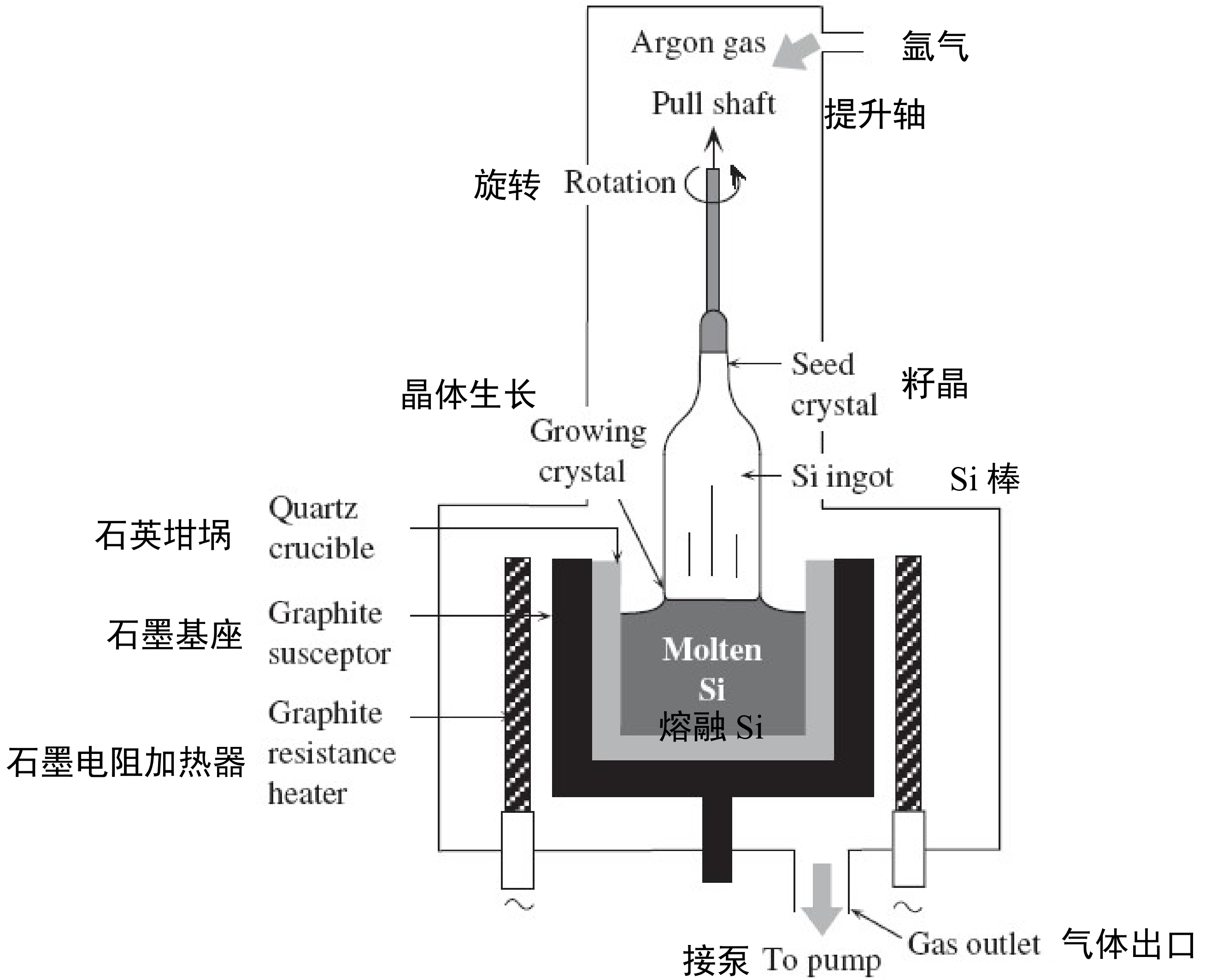
Similarities

localized distortion occurs in the lattice structure around the dislocation line, and only a small shear force can make the dislocation slip; The direction of the shear force is parallel to the direction of the slip.

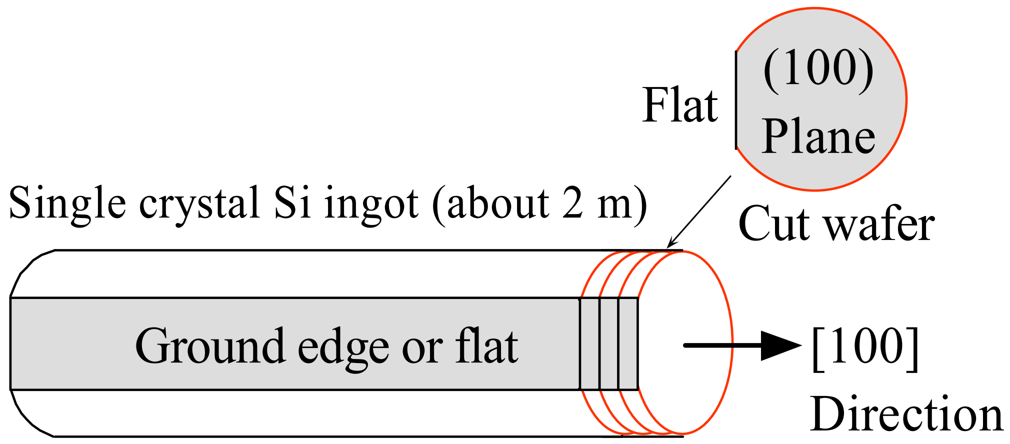
Differences

- in the edge dislocation, the dislocation line is perpendicular to the slip direction
- in the screw dislocation, the dislocation line is parallel to the slip direction

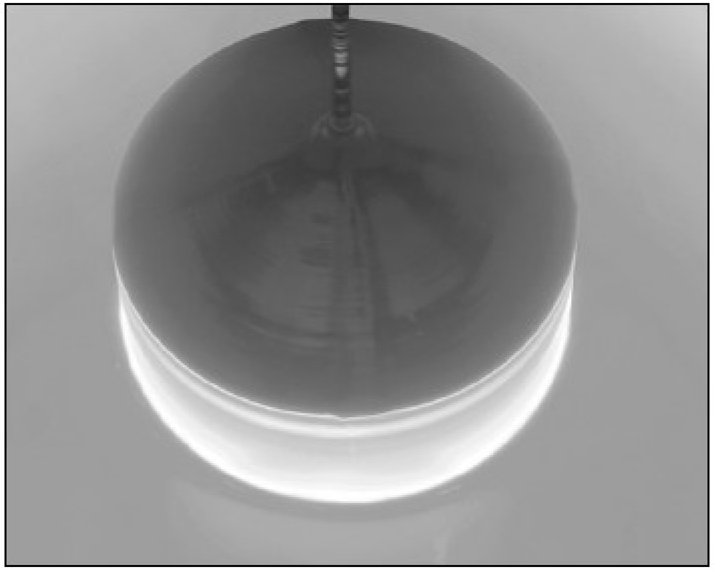
1.10 Single-crystal Czochralski growth 单晶的切克劳斯基生长法

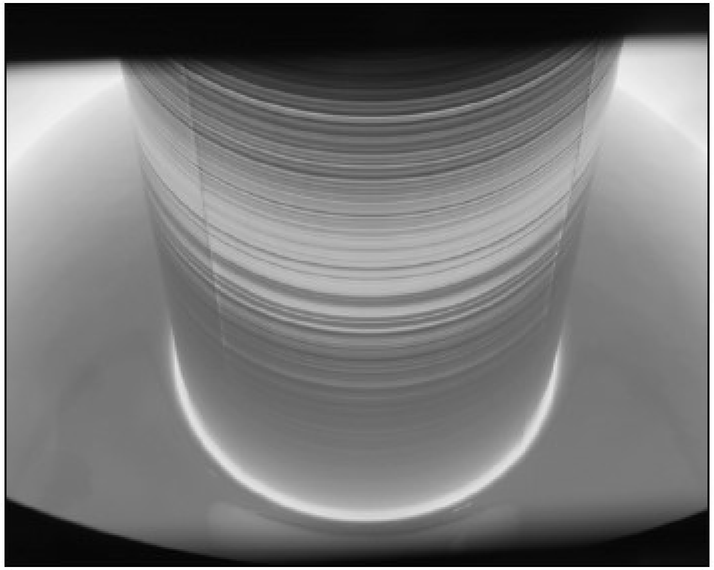


1.10 Single-crystal Czochralski growth 单晶的切克劳斯基生长法



(b) The crystallographic orientation of the silicon ingot is marked by grounding a flat. The ingot can be as long as 2 m. Wafers are cut using a rotating annular diamond saw. Typical wafer thickness is 0.6-0.7mm

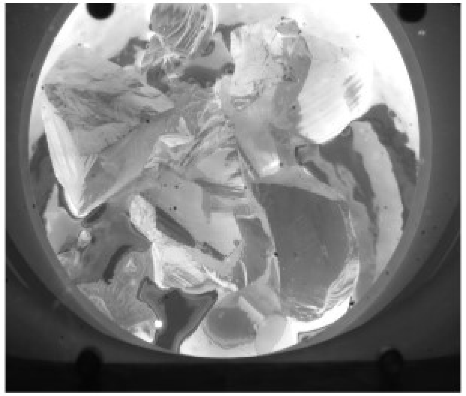


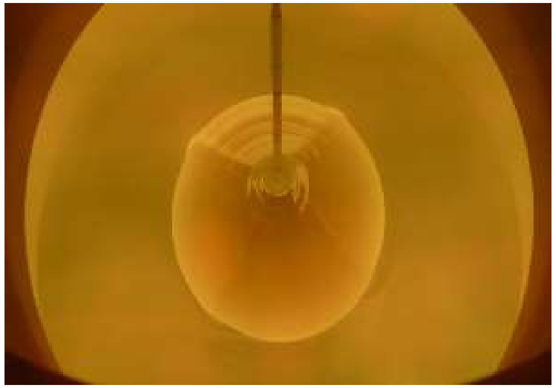


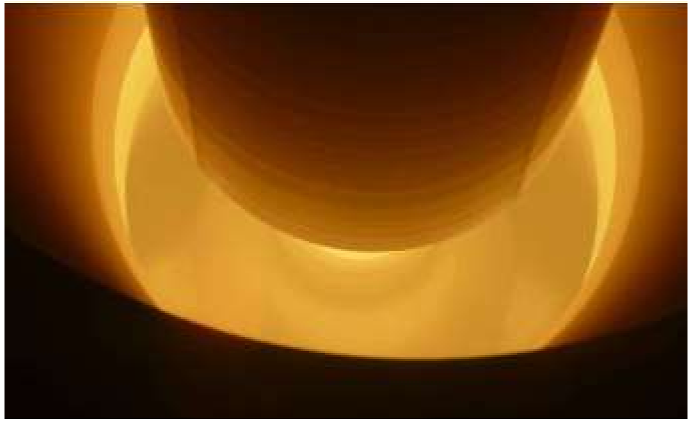
Silicon ingot being pulled from the melt in a Czochralski crystal drawer.

SOURCE: Courtesy of MEMC Electronic Materials, Inc.

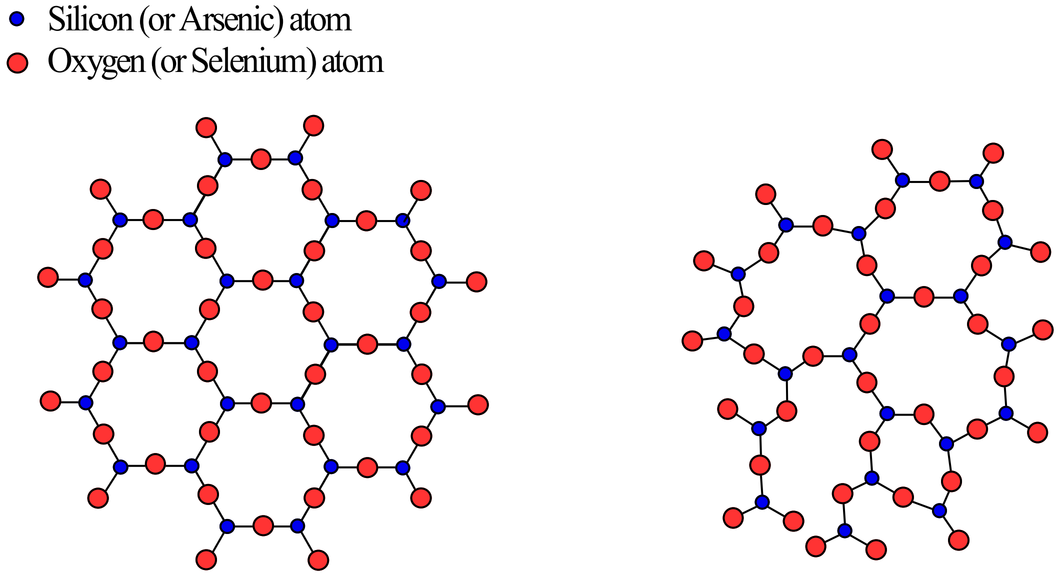
Procedures







1.11 Glasses and amorphous solids
玻璃和非晶态半导体



(a) A crystalline solid reminiscent to crystalline SiO_2.(Density = 2.6 g cm^{-3})

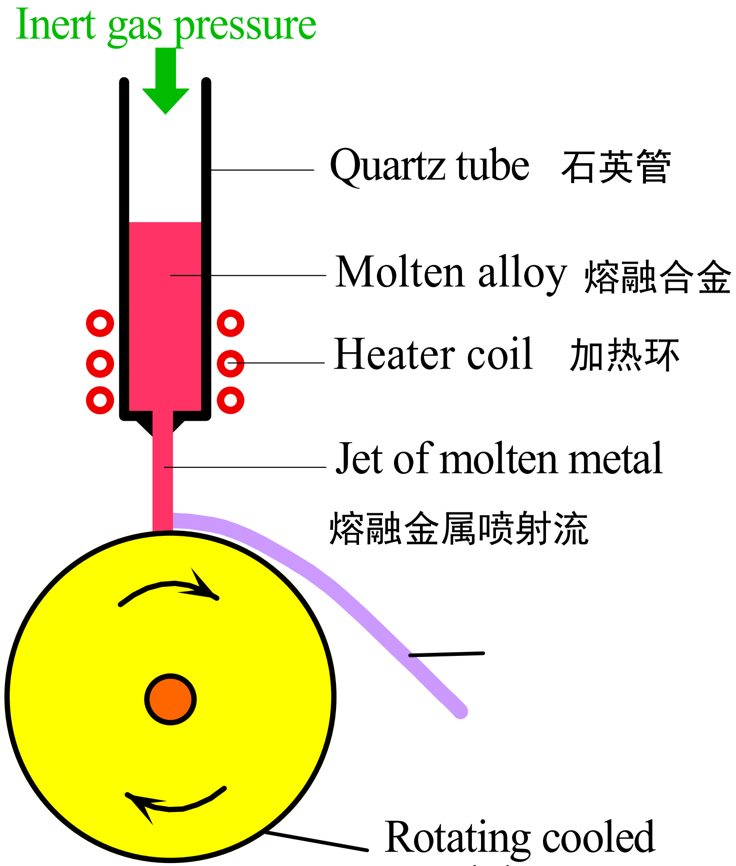
(b) An amorphous solid reminiscent to vitreous silica (SiO_2) cooled from the melt (Density = 2.2 g cm^{-3})

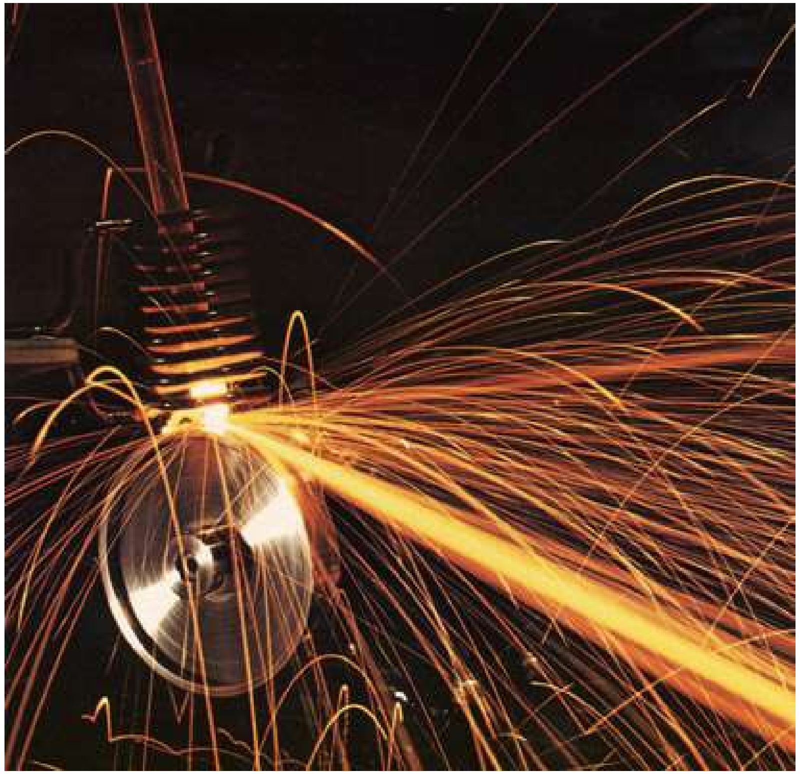
Crystalline and amorphous structures illustrated schematically in two dimensions.

Vitreous silica (石英玻璃) is a frozen liquid or supercooled liquid

Glass: Many amorphous solids are formed by rapidly cooling or quenching the liquid to temperatures where the atomic motions are so sluggish the crystallization is virtually halted.

As a consequence of the lack of long-range order, amorphous materials do not possess such crystalline imperfections as grain boundaries and dislocations.





Melt spinning involves squirting a jet of molten metal onto a rotating cool metal drum. The molten jet is instantly solidified into a glassy metal ribbon which is a few microns in thickness. The process produces roughly 1 to 2 kilometers of ribbon per minute.

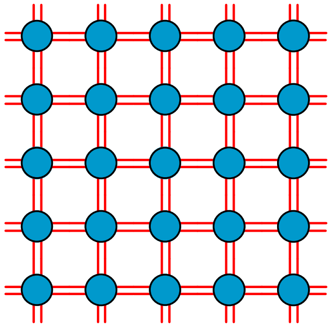
| SOURCE: Photo courtesy of the Estate of Fritz Goro.

It is possible to rapidly quench a molten metallic alloy, thereby bypassing crystallization, and forming a glassy metal commonly called a metallic glass. The process is called *melt spinning*.

通过快速冷却熔融金属合金，可能不结晶而直接形成金属玻璃。这一过程称为熔融纺丝。

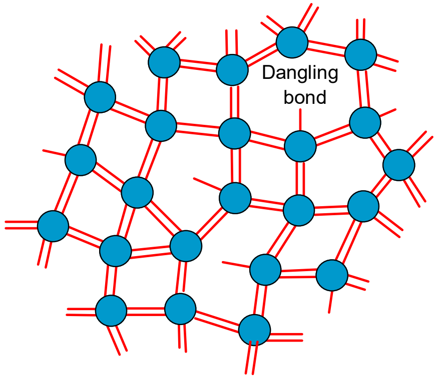
熔融纺丝就是将熔融的金属喷射在旋转的冷金属鼓上，迅速固化成几个微米厚的金属带。

1.11.2 Crystalline and amorphous silicon 单晶硅与非晶硅



(a) Two dimensional schematic representation of a silicon crystal

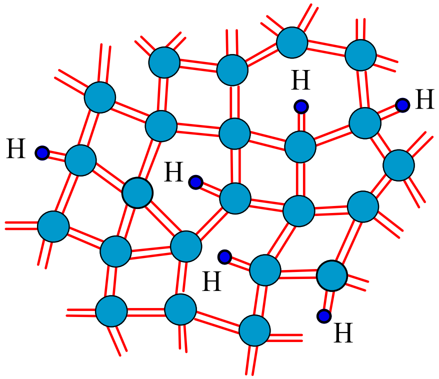
Si 晶体结构的二维图示



(b) Two dimensional schematic representation of the structure of amorphous silicon. The structure has voids and dangling bonds and there is no long range order.

非晶硅结构的二维图示，结构中包含孔与悬挂键，不存在长城有序

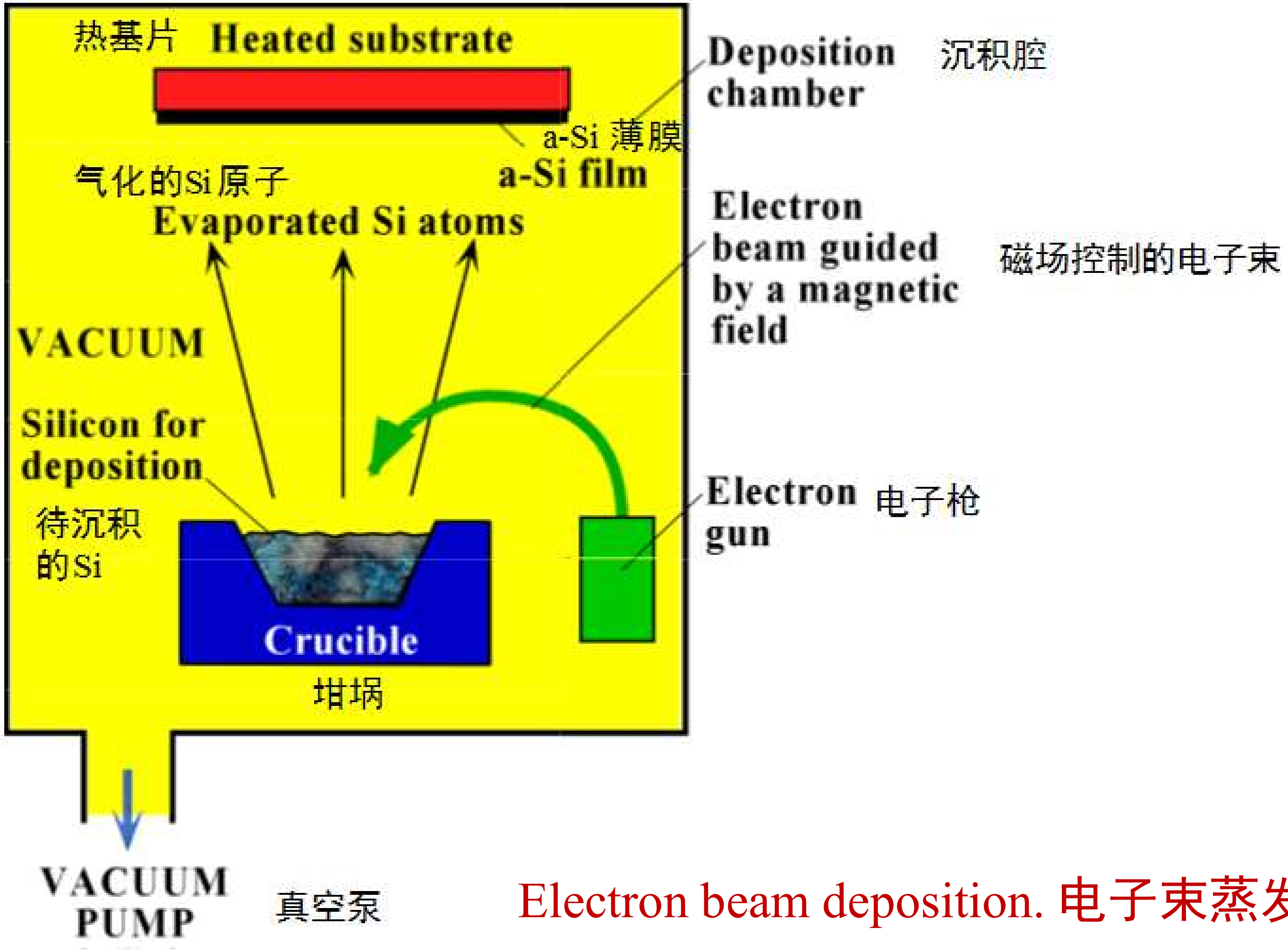
太阳能电池中的 *a*-Si: H



(c) Two dimensional schematic representation of the structure of hydrogenated amorphous silicon. The number of hydrogen atoms shown is exaggerated.

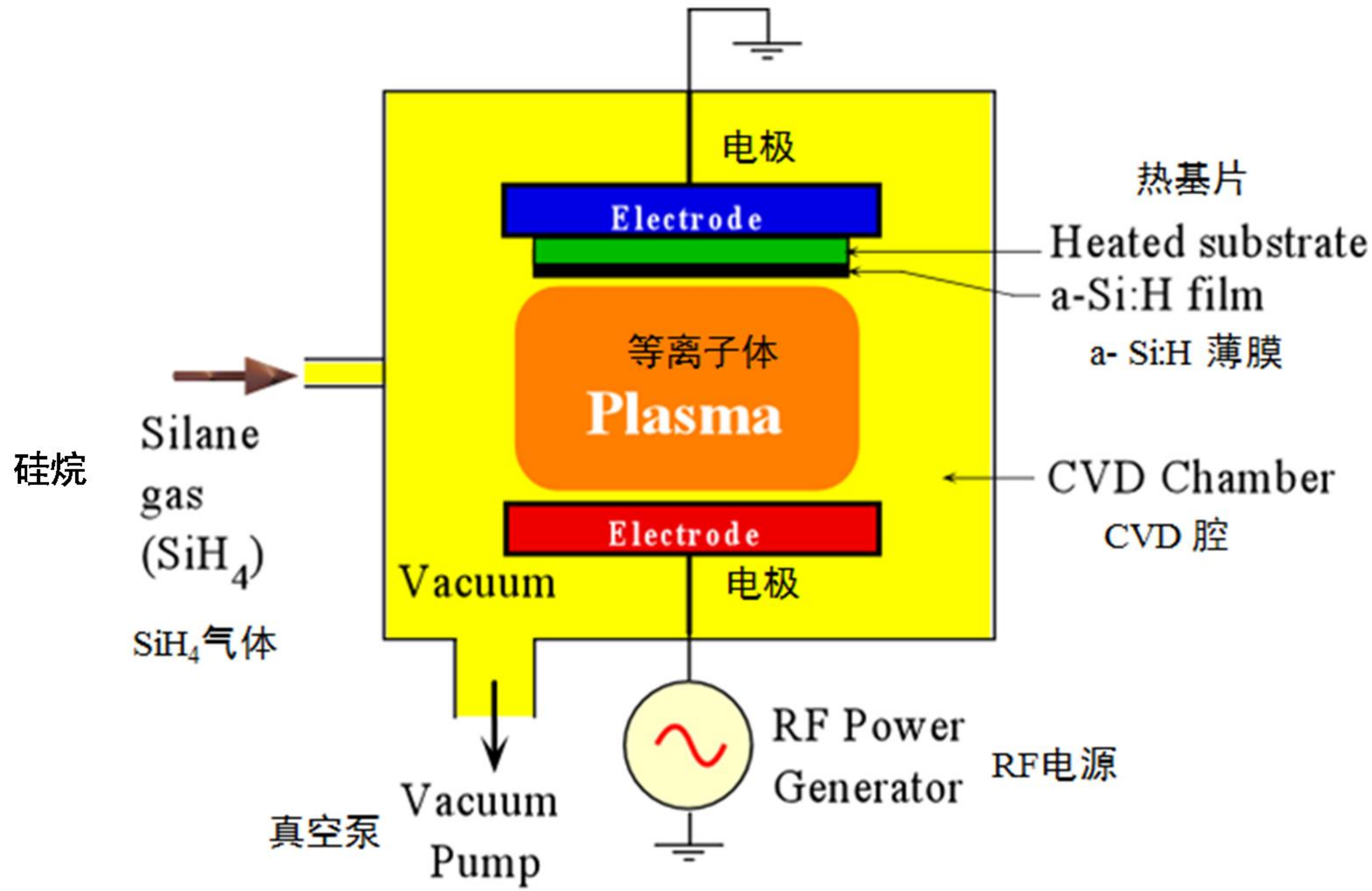
氢化无定形硅结构的二维图示，氢原子的数目大于实际情况

Silicon can be grown as a semiconductor crystal or as an amorphous semiconductor film. Each line represents an electron in a bond. A full covalent bond has two lines and a broken bond has one line.



Electron beam deposition. 电子束蒸发法

Amorphous silicon, a-Si, can be prepared by an electron beam evaporation of silicon. Silicon has a high melting temperature so that an energetic electron beam is used to melt the crystal in the crucible locally and thereby vaporize Si atoms. Si atoms condense on a substrate placed above the crucible to form a film of *a*-Si.



Hydrogenated amorphous silicon, a-Si:H, is generally prepared by the decomposition of silane molecules in a radio frequency (RF) plasma discharge. Si and H atoms condense on a substrate to form a film of a-Si:H.

Plasma Enhanced Chemical Vapor Deposition (PECVD)

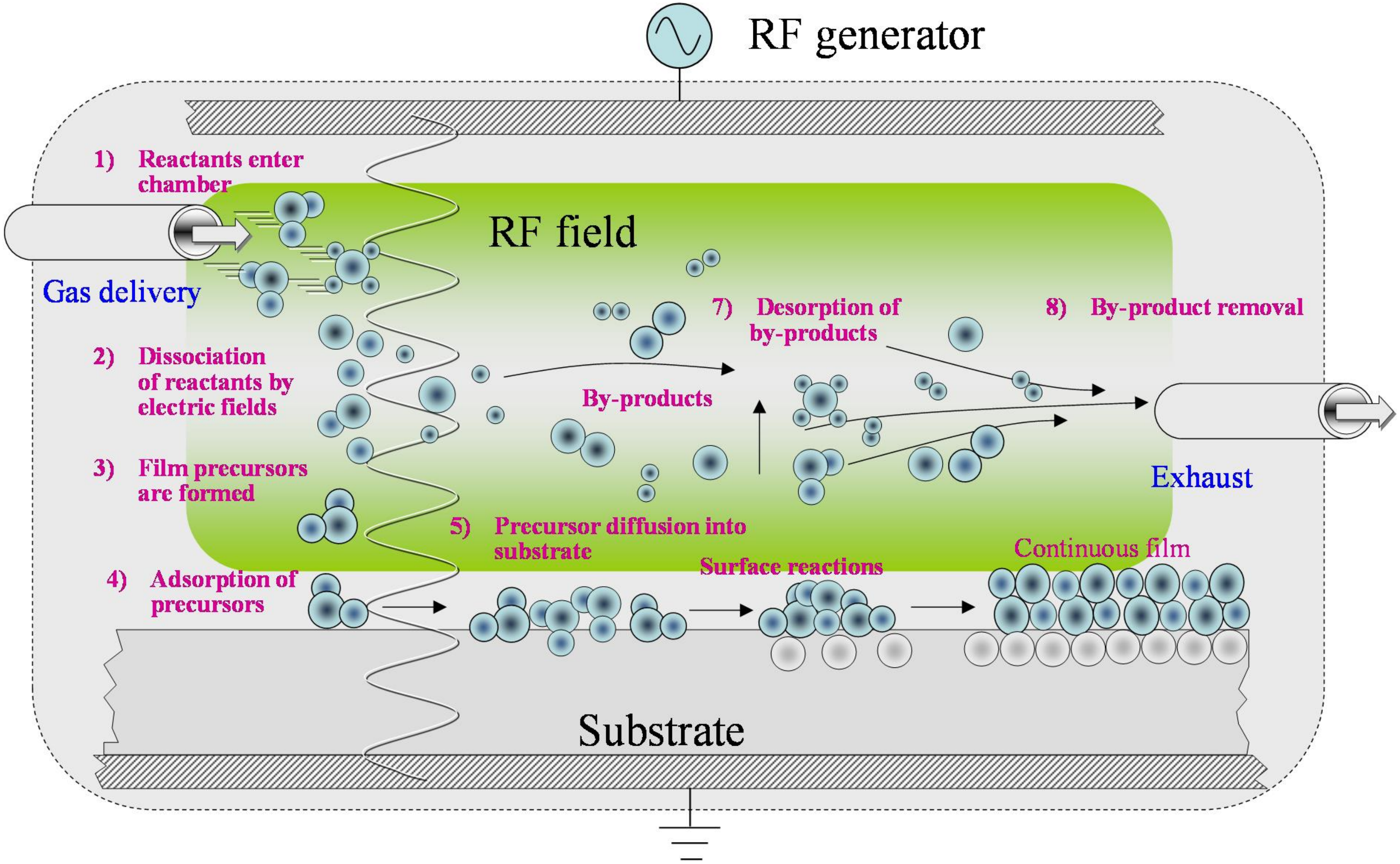


Table 1.5 summarizes the properties of crystalline and amorphous silicon, in terms of structure and applications.

Table 1.5 Crystalline and amorphous silicon

	Crystalline Si (c-Si)	Amorphous Si (a-Si)	Hydrogenated a-Si (a-Si:H)
Structure	Diamond cubic.	Short-range order only. On average, each Si covalently bonds with four Si atoms. Has microvoids and dangling bonds.	Short-range order only. Structure typically contains 10% H. Hydrogen atoms passivate dangling bonds and relieve strain from bonds.
Typical preparation	Czochralski technique.	Electron beam evaporation of Si.	Chemical vapor deposition of silane gas by RF plasma.
Density (g cm^{-3})	2.33	About 3–10% less dense.	About 1–3% less dense.
Electronic applications	Discrete and integrated electronic devices.	None	Large-area electronic devices such as solar cells, flat panel displays, and some photoconductor drums used in photocopying.